2024 CpSc Q7

Section: Software Design and Development

Topic: Record Structures — Array of Records and Search Algorithm

Question Summary

(a) Declare a record structure for Recipe and create a variable to hold data for 750 recipes.
(b) Write code to search the array for recipes where ingredient matches a given string and minutes ≤ a given limit, then display matching titles and cooking times.

Worked Solution

(a) Record declaration and array creation
Example in Python-style pseudocode:

```
class Recipe:
    def __init__(self, title, ingredient, minutes, cost, avgRating):
        self.title = title
        self.ingredient = ingredient
        self.minutes = minutes
        self.cost = cost
        self.avgRating = avgRating

# Array (list) to hold 750 Recipe records
recipes = [Recipe('', '', 0, 0.0, 0.0) for i in range(750)]
```

Other languages may use STRUCT ... END STRUCT or RECORD ... END RECORD. The key idea is a record defining 5 fields, and an array that can hold 750 such records.

(b) Search code (Python-style pseudocode)

```
def searchRecipes(recipes, targetIngredient, maxTime):
    found = False
    for r in recipes:
        if r.ingredient == targetIngredient and r.minutes <=
maxTime:
            print(r.title, 'requiring', r.minutes, 'minutes.')
            found = True
    if not found:
        print('No matches found.')
```

This uses a simple linear search. The found flag ensures that if no recipes match, the message ‘No matches found.’ is printed only once.

Alternative (pseudocode)

```
RECORD Recipe IS {STRING title, STRING ingredient,
INTEGER minutes, REAL cost, REAL avgRating}
DECLARE recipes : ARRAY[1..750] OF Recipe

SET found ← FALSE
FOR i FROM 1 TO 750 DO
    IF recipes[i].ingredient = targetIngredient AND
recipes[i].minutes ≤ maxTime THEN
        OUTPUT recipes[i].title & ' requiring ' &
recipes[i].minutes & ' minutes.'
        SET found ← TRUE
    ENDIF
ENDFOR
IF found = FALSE THEN OUTPUT 'No matches found.'
```

Final Answer

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(a) Define a record for Recipe and create an array/list of 750 Recipe elements.
(b) Linear search: test each recipe for ingredient match and cooking time ≤ limit; display matches, otherwise ‘No matches found.’

Revision Tips

- A record (or struct/class) groups multiple fields for each item — ideal for storing recipe data.
- An array of records allows structured, indexed storage of many similar items (750 here).
- Use a Boolean found flag when you must print a single ‘no match’ message after a search.
- Test both conditions (ingredient and minutes) with logical AND.
- For large data sets, an indexed or binary search could improve efficiency — but a linear search is sufficient here.

Exam Alignment

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Matches 2024 MI: record correctly defined with 5 fields; array for 750 records; search algorithm checks both ingredient and time; displays recipe title and time; outputs ‘No matches found.’ if none match.